Core stability does not imply egalitarian stability: a sharp six-player counterexample

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Abstract

Dietzenbacher (2021) asked whether every transferable-utility game whose core is a von Neumann–Morgenstern stable set is egalitarian stable, meaning that the procedural egalitarian solution selects a core allocation. We answer no by applying a player-specific additive perturbation to a classical six-player stable-core game. The perturbation translates the core and preserves domination, exactness, monotonicity, and superadditivity, but its egalitarian claims sum to 13/2, exceeding the grand-coalition worth 6. We give a self-contained stability proof: five tight pairs reduce every imputation not dominated by the core to a one-parameter discrepancy, and one of two tight triples eliminates either sign. The player bound is sharp, because core stability and core largeness coincide for at most five players, and large-core games are egalitarian stable. The construction also shows that egalitarian stability is not invariant under player-specific additive transformations. The exact-game half of Dietzenbacher's question had already been exhibited by Mauri Masdeu (2017).

Keywords: transferable-utility games; stable core; stable set; procedural egalitarian solution; exact games.

1 Introduction

The procedural egalitarian solution extends the constrained egalitarian solution of Dutta and Ray (1989) beyond convex games (Dietzenbacher et al. 2017). Its natural domain is the class of *egalitarian stable* games: those on which the procedure selects from the core (Dietzenbacher 2021). This class contains all games with a large core. Two familiar classes therefore supplied unresolved boundary questions. Dietzenbacher (2021) asked whether every exact game is egalitarian stable and whether the same is true whenever the core is a von Neumann–Morgenstern stable set.

The two halves have different prior-art histories. Mauri Masdeu (2017, Example 5) applied the relevant maximum-average procedure to the five-player exact game of Biswas et al. (1999), obtained the zero vector, and observed that it lies outside the core. In the later terminology of Dietzenbacher (2021), that example is not egalitarian stable. We recall this point in Section 3 but claim no novelty for the exact-game half.

*Online Resource 1 contains machine-readable witness data and standalone exact-arithmetic verifiers.

For the stable-core half, we start from a classical six-player orthogonal game. Six-player stable-but-not-large examples go back at least to Biswas et al. (2001); the precise segment-core game used here is recorded by Shellshear and Sudhölter (2009, Example 4.3). We add the nonuniform additive vector $c = (1, 0, 0, 0, 1, 0)$. After a permutation of the classical game, the perturbed game is generated by

$$A = (1, 1, 0, 1, 3, 0), \qquad B = (2, 0, 1, 0, 1, 2), \qquad v(S) = \min\{A(S), B(S)\}.$$

Its core is exactly the segment $[A, B]$, and its egalitarian claims are

$$\widehat{\gamma} = \left(\tfrac{3}{2}, \tfrac{1}{2}, 1, \tfrac{1}{2}, \tfrac{3}{2}, \tfrac{3}{2}\right), \qquad \widehat{\gamma}(N) = \tfrac{13}{2} > 6 = v(N).$$

Thus the perturbed game is not egalitarian stable. Additive transformations translate imputations and cores and preserve the dominance relation, so its stable core is inherited from the classical game. We also give a short, self-contained proof in the perturbed coordinates: five tight pairs and two tight triples suffice to dominate every imputation outside the segment. The new ingredient is therefore not the underlying core geometry, but the fact that this particular perturbation preserves that geometry while destroying egalitarian stability.

The result is sharp. Estévez-Fernández (2012) proved that core stability and core largeness coincide for games with at most five players, while large-core games are egalitarian stable (Dietzenbacher 2021). Hence six is the first possible number of players. The example also shows that the inclusion of large-core games in stable-core games is strict even after intersecting with the class of exact, monotone, superadditive games.

Core stability has a general finite characterization through nested balancedness (Grabisch and Sudhölter 2024). Our main proof does not require that machinery. For reproducibility, however, Appendix A gives a specialized exact arrangement test for segment cores. A standalone verifier checks all 20,349 candidate bases in rational arithmetic and independently confirms the hand proof.

2 Preliminaries

Let $N = \{1, \dots, n\}$. A transferable-utility game is a function $v\colon 2^N \to \mathbb{R}$ with $v(\emptyset) = 0$. For $x \in \mathbb{R}^N$ and $S \subseteq N$, write $x(S) = \sum_{i \in S} x_i$. The core is

$$\mathrm{Core}(v) = \{x \in \mathbb{R}^N : x(N) = v(N),\ x(S) \geq v(S) \text{ for every } S \subseteq N\}.$$

An *imputation* is an efficient and individually rational vector: $x(N) = v(N)$ and $x_i \geq v(\{i\})$ for every i. Given imputations x, y, x *dominates* y via a nonempty coalition S if

$$x_i > y_i \quad (i \in S), \qquad x(S) \leq v(S).$$

The core is internally stable. It is a von Neumann–Morgenstern *stable set* precisely when it is externally stable: every imputation outside the core is dominated by a core element (von Neumann and Morgenstern 1944). We then say that v has a *stable core*.

Following Schmeidler (1972), a game is *exact* if every coalition constraint is attained in the core:

$$v(S) = \min_{x \in \mathrm{Core}(v)} x(S) \qquad (S \subseteq N).$$

The core is *large* if every vector in the upper core, $\{y : y(S) \geq v(S)\ \forall S\}$, lies above some core point (Sharkey 1982). Every large-core game has a stable core and is egalitarian stable (Sharkey 1982; Dietzenbacher 2021).

2.1 The procedural egalitarian claims

We recall only the part of the procedural egalitarian solution needed below (Dietzenbacher et al. 2017; Dietzenbacher 2021). Let $P^0 = \emptyset$. At round k, claims γ_i have already been fixed for $i \in P^{k-1}$. For each nonempty $S \subseteq N$, define

$$\chi_i^k(S) = \begin{cases} \gamma_i, & i \in S \cap P^{k-1}, \\ \dfrac{v(S) - \sum_{j \in S \cap P^{k-1}} \gamma_j}{|S \setminus P^{k-1}|}, & i \in S \setminus P^{k-1}, \end{cases}$$

with the first line applying to every member when $S \subseteq P^{k-1}$. Let $m_i^k = \max_{S \ni i} \chi_i^k(S)$. The round-$k$ admissible collection is

$$\mathcal{A}^k = \left\{ S \neq \emptyset : \sum_{i \in S} \chi_i^k(S) = v(S) \text{ and } \chi_i^k(S) = m_i^k \text{ for all } i \in S \right\}.$$

The players in $\bigcup_{S \in \mathcal{A}^k} S$ are added to P^k, and each newly fixed player i receives claim $\gamma_i = m_i^k$. The procedure terminates with the *egalitarian claims* vector $\widehat{\gamma}$. A game is *egalitarian stable* (or PES-stable) if the grand coalition belongs to the terminal admissible collection. On exactly this class, the procedural egalitarian solution selects a core allocation (Dietzenbacher 2021).

Lemma 2.1. *If $\widehat{\gamma}(N) > v(N)$, then v is not egalitarian stable.*

Proof. At the terminal round every player is fixed, so $\chi_i(N) = \widehat{\gamma}_i$ for all i. Membership of N in the terminal admissible collection requires $\sum_i \chi_i(N) = v(N)$, which fails when $\widehat{\gamma}(N) > v(N)$. □

2.2 A lower-envelope generator

Both the prior-work observation and the new witness use the following standard finite form of the lower-envelope description of exact games.

Lemma 2.2 (finite lower envelopes)**.** *Let $Q \subset \mathbb{R}^N$ be a nonempty finite set whose vectors have common sum τ, and define*

$$v(S) = \min_{q \in Q} q(S).$$

Then $v(N) = \tau$, every $q \in Q$ lies in Core(v)*, and v is exact. If all coordinates of the generators are nonnegative, then v is monotone and superadditive.*

Proof. For each $q \in Q$, $q(N) = \tau = v(N)$ and $q(S) \geq \min_{p \in Q} p(S) = v(S)$, so $q \in$ Core(v). Given S, choose a generator q^S attaining the minimum. Then $q^S(S) = v(S)$, proving exactness.

If the generators are nonnegative, each $q(S)$ is monotone in S, hence so is their pointwise minimum. If S and T are disjoint, then every $q \in Q$ satisfies

$$q(S \cup T) = q(S) + q(T) \geq v(S) + v(T).$$

Taking the minimum over q proves superadditivity. □

3 The exact-game half is prior work

The exact-game half of Dietzenbacher (2021) requires no new counterexample. For five players, Mauri Masdeu (2017, Example 5) considers the exact game of Biswas et al. (1999)

$$a = (1, -1, -1, 0, 0), \qquad b = (0, 1, 0, -1, -1), \qquad u(S) = \min\{a(S), b(S)\}.$$

Both generators sum to -1, while the maximum coalition average is 0. Mauri Masdeu computes the maximizing collection

$$\{1\}, \quad \{1, 2\}, \quad \{1, 3\}, \quad \{1, 2, 4\}, \quad \{1, 2, 5\},$$

which covers all players, and obtains the vector $(0, 0, 0, 0, 0)$ outside the core. In the terminology recalled in Section 2, these are exactly the first-round admissible coalitions and terminal egalitarian claims. Since $\widehat{\gamma}(N) = 0 > -1 = u(N)$, Lemma 2.1 says that the game is not egalitarian stable. Thus both the construction and the decisive egalitarian computation predate the 2021 question; this paper makes no novelty claim for that half.

4 A perturbation of a classical stable-core game

We first record what an additive transformation preserves. For $c \in \mathbb{R}^N$, write $c(S) = \sum_{i \in S} c_i$.

Lemma 4.1 (additive transformations)**.** *If $v(S) = w(S) + c(S)$ for every $S \subseteq N$, then*

$$I(v) = I(w) + c, \qquad \mathrm{Core}(v) = \mathrm{Core}(w) + c,$$

where I denotes the imputation set. Moreover, v is exact if and only if w is exact, and v has a stable core if and only if w does.

Proof. For every $x \in \mathbb{R}^N$ and $S \subseteq N$,

$$(x + c)(S) \geq v(S) \quad \Longleftrightarrow \quad x(S) \geq w(S),$$

and efficiency and individual rationality transform in the same way. This proves the two translations and the equivalence of exactness. Finally, $x + c$ dominates $y + c$ through S in v if and only if x dominates y through S in w: the strict coordinate comparisons cancel c, as does the affordability inequality. External stability is therefore preserved. □

Let $N = \{1, \ldots, 6\}$ and consider

$$U = (0, 1, 0, 1, 2, 0), \qquad V = (1, 0, 1, 0, 0, 2), \qquad w(S) = \min\{U(S), V(S)\}.$$

Under the coordinate order $(5, 2, 4, 6, 1, 3)$, its generators become

$$(2, 1, 1, 0, 0, 0) \quad \text{and} \quad (0, 0, 0, 2, 1, 1).$$

This is the classical exact stable-core orthogonal game recorded by Shellshear and Sudhölter (2009, Example 4.3); its core is the segment between the two generators. We use the nonuniform additive vector $c = (1, 0, 0, 0, 1, 0)$. The perturbed game $v = w + c$ has generators

$$A = U + c = (1, 1, 0, 1, 3, 0), \qquad B = V + c = (2, 0, 1, 0, 1, 2),$$

and hence

$$v(S) = \min\{A(S), B(S)\}. \tag{1}$$

Both generators sum to 6 and are nonnegative. By Lemma 2.2, v is exact, monotone, and superadditive. Lemma 4.1 and the cited classical example already imply that its core is stable. The next two sections give a self-contained segment description and stability proof in the perturbed coordinates.

4.1 The core is a segment

Put

$$x_t = (1-t)A + tB = (1+t, 1-t, t, 1-t, 3-2t, 2t), \qquad 0 \le t \le 1. \tag{2}$$

Proposition 4.2. *The core of v is exactly $\{x_t : 0 \le t \le 1\}$.*

Proof. Lemma 2.2 puts A and B in the core, so convexity of the core gives $x_t \in \mathrm{Core}(v)$ for $0 \le t \le 1$.

Conversely, let $y \in \mathrm{Core}(v)$. The three disjoint pair constraints

$$y_1 + y_2 \ge 2, \qquad y_3 + y_4 \ge 1, \qquad y_5 + y_6 \ge 3$$

sum to $y(N) \ge 6 = v(N)$, so efficiency makes all three equalities. Individual rationality then permits the parametrization

$$y = (1+t, 1-t, s, 1-s, 3-u, u), \qquad t, s \in [0,1], \quad u \in [0,2].$$

The fully tight pair constraints for $\{2,3\}$ and $\{1,4\}$ give, respectively,

$$1 - t + s \ge 1, \qquad 2 + t - s \ge 2,$$

and hence $s = t$. Finally, the fully tight triples $\{1,3,5\}$ and $\{2,4,6\}$ have worths 4 and 2. Their core inequalities become

$$4 + 2t - u \ge 4, \qquad 2 - 2t + u \ge 2,$$

so $u = 2t$. Therefore $y = x_t$. □

4.2 The egalitarian procedure leaves the core

The full coalition function is already specified by Equation (1); its 63 nonempty values can therefore be checked without a separate table. Direct evaluation of the procedure gives the two-round trace in Table 1. In the first round, the maximum equal shares for players $1, \dots, 6$ are

$$\left(\tfrac{3}{2}, \tfrac{5}{4}, \tfrac{4}{3}, \tfrac{5}{4}, \tfrac{3}{2}, \tfrac{3}{2}\right).$$

Only $\{1,5\}$ and $\{5,6\}$ simultaneously maximize the shares of all their members, so players $1, 5, 6$ are fixed at $3/2$. The second round fixes the remaining players.

Thus the terminal claims are

$$\widehat{\gamma} = \left(\tfrac{3}{2}, \tfrac{1}{2}, 1, \tfrac{1}{2}, \tfrac{3}{2}, \tfrac{3}{2}\right), \qquad \widehat{\gamma}(N) = \tfrac{13}{2} > 6 = v(N). \tag{3}$$

By Lemma 2.1, v is not egalitarian stable. The supplement recomputes the trace from Equation (1) using exact rational arithmetic.

Table 1: Procedural egalitarian trace for the perturbed six-player game; a digit string denotes the corresponding coalition.

round	admissible collection $\mathcal{A}^k$	newly fixed	new claims
1	15, 56	$1, 5, 6$	$\frac{3}{2}, \frac{3}{2}, \frac{3}{2}$
2	12, 14, 15, 135, 56, 1256, 1456	$2, 3, 4$	$\frac{1}{2}, 1, \frac{1}{2}$

Corollary 4.3. *Egalitarian stability is not invariant under player-specific additive transformations, even among exact games with stable cores.*

Proof. For the unperturbed game w, the first round fixes players 5 and 6 at 1 through $\{5, 6\}$. The second round fixes the other players at $1/2$ and admits the grand coalition. Thus w is egalitarian stable, with claims

$$\left(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, 1, 1\right) = \tfrac{1}{2}(U + V) \in \mathrm{Core}(w).$$

The additively transformed game $v = w + c$ is not egalitarian stable by Equation (3). □

5 The core is externally stable

The segment in Equation (2) has five useful two-player projections:

pair P	$v(P)$	$x_t\|_P$
$\{1, 2\}$	2	$(1 + t, 1 - t)$
$\{3, 4\}$	1	$(t, 1 - t)$
$\{5, 6\}$	3	$(3 - 2t, 2t)$
$\{2, 3\}$	1	$(1 - t, t)$
$\{1, 4\}$	2	$(1 + t, 1 - t)$

In each row, as t ranges over $[0, 1]$, the projection sweeps the entire two-player imputation segment for the corresponding subgame.

Lemma 5.1 (pair domination)**.** *Let P be one of the five pairs in the table. If an imputation y satisfies $y(P) < v(P)$, then some x_t dominates y via P.*

Proof. Write $P = \{i, j\}$ and $\varepsilon = v(P) - y(P) > 0$. The vector

$$z = (y_i + \varepsilon/2,\ y_j + \varepsilon/2)$$

is individually rational for the two-player subgame and sums to $v(P)$. Hence it lies on the full segment swept by $x_t|_P$, so $z = x_t|_P$ for some $t \in [0, 1]$. Both members of P strictly gain, and $x_t(P) = v(P)$; thus x_t dominates y via P. □

Theorem 5.2. *The game in Equation (1) is an exact, monotone, superadditive six-player game with a stable core that is not egalitarian stable.*

Proof. Exactness, monotonicity, and superadditivity follow from Lemma 2.2; non-egalitarian-stability follows from Equation (3). It remains to prove external stability.

Let y be an imputation that is not dominated by any core element. Lemma 5.1 implies

$$y_1 + y_2 \geq 2, \quad y_3 + y_4 \geq 1, \quad y_5 + y_6 \geq 3, \qquad y_2 + y_3 \geq 1, \quad y_1 + y_4 \geq 2.$$

Efficiency makes the first three inequalities equalities. As in the proof of Proposition 4.2, individual rationality and the last two inequalities then give

$$y = (1 + t, 1 - t, t, 1 - t, 3 - u, u), \qquad t \in [0, 1], \quad u \in [0, 2].$$

Suppose first that $u > 2t$. Choose r strictly between t and $u/2$. The bounds on t and u ensure $r \in [0, 1]$, and

$$x_r|_{\{1,3,5\}} = (1 + r, r, 3 - 2r) > (1 + t, t, 3 - u) = y|_{\{1,3,5\}}.$$

Since $x_r(\{1, 3, 5\}) = 4 = v(\{1, 3, 5\})$, x_r dominates y through that triple, a contradiction.

If $u < 2t$, choose r strictly between $u/2$ and t. Then

$$x_r|_{\{2,4,6\}} = (1 - r, 1 - r, 2r) > (1 - t, 1 - t, u) = y|_{\{2,4,6\}},$$

and $x_r(\{2, 4, 6\}) = 2 = v(\{2, 4, 6\})$, again a contradiction. Consequently $u = 2t$, so $y = x_t \in \mathrm{Core}(v)$ by Proposition 4.2. Every imputation outside the core is therefore dominated by a core element. □

Corollary 5.3 (sharpness)**.** *Six is the smallest number of players for which a game can have a stable core without being egalitarian stable.*

Proof. Theorem 5.2 supplies a six-player game. For at most five players, core stability is equivalent to core largeness (Estévez-Fernández 2012), and every large-core game is egalitarian stable (Dietzenbacher 2021). □

6 Discussion

The counterexample clarifies the relation among three notions:

$$\{\text{large-core games}\} \subsetneq \{\text{stable-core games}\},$$
$$\{\text{stable-core games}\} \not\subseteq \{\text{egalitarian-stable games}\}.$$

The second noninclusion is witnessed within exact, monotone, superadditive games. The underlying segment geometry is classical; what changes is the procedural outcome under a nonuniform additive perturbation. This is precisely where the two stability notions diverge: core stability is invariant under additive transformations, whereas Corollary 4.3 shows that egalitarian stability is not.

The mechanism behind stability is local. Any imputation that underpays one of five pairs is immediately dominated on that pair. Avoiding those pair dominances forces all but one degree of freedom to agree with the core parameter. The two triples then point in opposite directions: one dominates when the residual coordinate is too large, the other when it is too small. By contrast, after the perturbation the egalitarian procedure combines different tight coalitions across its two rounds and produces mutually unaffordable claims. The example therefore shows that external dominance power of the core does not control the global compatibility of procedural claims.

A A specialized exact cross-check for segment cores

Theorem 5.2 has a short direct proof. This appendix records a general-purpose cross-check used during the audit. It is specialized to a segment core and complements the general nested-balancedness characterization of Grabisch and Sudhölter (2024).

Suppose $\mathrm{Core}(v) = [A, B]$ and write $d = B - A$ and $x_t = A + td$ for $0 \le t \le 1$. Let

$$E = \{y : y(N) = v(N)\}, \qquad I = \{y \in E : y_i \ge v(\{i\})\ \forall i\}.$$

For a coalition S, let $F_S = \{x \in [A, B] : x(S) = v(S)\}$. Since $x_t(S) - v(S)$ is affine and nonnegative on $[0, 1]$, F_S is empty, $\{A\}$, $\{B\}$, or the entire segment. A core element that dominates through S must belong to F_S, because core feasibility gives $x(S) \ge v(S)$ while domination requires $x(S) \le v(S)$.

When $F_S = [A, B]$, put

$$P_S = \{i \in S : d_i > 0\}, \quad M_S = \{j \in S : d_j < 0\}, \quad Z_S = \{k \in S : d_k = 0\},$$

and, for $i \in P_S$, $j \in M_S$, define

$$h_{ij}(y) = d_j(y_i - A_i) - d_i(y_j - A_j).$$

Lemma A.1. *If $F_S = [A, B]$, then a point $y \in I$ is dominated through S by a point of the segment if and only if*

$$\begin{aligned} y_k &< A_k \quad (k \in Z_S), \\ y_i &< B_i \quad (i \in P_S), \\ y_j &< A_j \quad (j \in M_S), \\ h_{ij}(y) &> 0 \quad (i \in P_S,\ j \in M_S). \end{aligned}$$

If $F_S = \{A\}$ or $\{B\}$, the corresponding condition is $y_i < A_i$ for all $i \in S$ or $y_i < B_i$ for all $i \in S$, respectively.

Proof. For $d_i \ne 0$, set $q_i = (y_i - A_i)/d_i$. The inequalities $x_{t,i} > y_i$ are $t > q_i$ when $d_i > 0$ and $t < q_i$ when $d_i < 0$; if $d_i = 0$ they are $y_i < A_i$. A parameter $t \in [0, 1]$ exists exactly when

$$q_i < 1\ (i \in P_S), \qquad q_j > 0\ (j \in M_S), \qquad q_i < q_j\ (i \in P_S, j \in M_S).$$

These are the displayed inequalities; multiplication by $d_i > 0 > d_j$ gives the stated sign for h_{ij}. The endpoint cases are immediate. □

Proposition A.2 (finite segment-core certificate)**.** *In the efficiency space E, form the hyperplane family*

$$\begin{aligned} \mathcal{H} = \{&y_i = v(\{i\}),\ y_i = A_i,\ y_i = B_i : i \in N\} \\ &\cup \{h_{ij}(y) = 0 : d_i > 0 > d_j\}. \end{aligned}$$

The core is stable if and only if every vertex of the $\mathcal{H}$-arrangement that lies in $I \setminus [A, B]$ is dominated by a point of $[A, B]$.

Proof. For each S, Lemma A.1 expresses the set D_S of imputations dominated through S as an intersection of open halfspaces whose boundaries belong to $\mathcal{H}$; the endpoint and empty-face cases have the same property. Hence $I \setminus D_S$ is a finite union of closed halfspaces bounded by $\mathcal{H}$. The undominated set

$$U = I \cap \bigcap_{\emptyset \neq S \subsetneq N} (I \setminus D_S)$$

is therefore a finite union of compact polytopes with facets supported by $\mathcal{H}$. Also $[A, B] \subseteq U$, because one core point cannot dominate another.

If $U \setminus [A, B]$ is nonempty, some polytope piece P of this union is not contained in $[A, B]$. If every vertex of P belonged to the convex segment $[A, B]$, then P, the convex hull of its vertices, would also be contained in the segment. Thus P has a vertex in $U \setminus [A, B]$. At that vertex, $n-1$ independent hyperplanes from $\mathcal{H}$, together with efficiency, determine the point, so it is an arrangement vertex. The converse is immediate. □

For the perturbed game, all six entries of d are nonzero, with three positive and three negative. The individual-rationality floor for each player duplicates one endpoint-coordinate plane. Thus $\mathcal{H}$ has $12 + 3 \cdot 3 = 21$ distinct hyperplanes in the five-dimensional efficiency space. The exact verifier enumerates all $\binom{21}{5} = 20{,}349$ candidate bases. It obtains 230 distinct intersections, of which 104 are imputations; exactly the two endpoints lie in the core, and each of the remaining 102 vertices has a rationally constructed dominator x_t. Proposition A.2 then independently certifies core stability. The supplement repeats the same check on an equivalent alternate normalization retained as a regression case.

B Verification and reproducibility

The main result is proved analytically in Theorem 5.2; computation is not part of its logical basis. Online Resource 1 supplies two independent standard-library Python verifiers and coalition data in exact rational form. `verify_paper2.py` reconstructs each game, recomputes the terminal egalitarian claims and admissible collection, checks the two-round trace of the main example, and verifies the generator representation and tight affine-line system. `verify_stability.py` implements the optional certificate of Appendix A. For the reported game and an equivalent alternate normalization retained as a regression check, it constructs the 21 relevant hyperplanes, solves all $\binom{21}{5} = 20{,}349$ bases using exact fractions, and checks an explicit segment-core dominator for each outside-core imputation vertex. The deterministic counts are 230 distinct intersections, 104 imputations, 2 core vertices, and 102 exactly dominated outside-core vertices in each normalization. No floating-point classification, external solver, or unpreserved checkpoint is used.

The perturbation was found computationally. The prior-art identification, final proofs, coalition data, and verification scripts were then checked independently in exact arithmetic. Large-language-model systems assisted with search, literature discovery, verification code, proof criticism, and drafting. All mathematical claims were reviewed by the author; the correction history and method of discovery are retained in the project provenance log.

Statements and Declarations

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